

Literature Report: Geopolitics, Hardware, and the Future of Astronomical Data

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Github Link (all documents): <https://github.com/noushinh/quantum-hardware-geopolitics>

Introduction: The Classical Dependency and the New "Chip War"

The modern global infrastructure - from financial institutions to hospital life support systems - is fundamentally dependent on a fragile, invisible engine: the classical silicon microchip. As highlighted in the symposium, these classical chips are tiny pieces of hyper-pure silicon packed with billions of transistors. However, the manufacturing of these components relies on a highly precarious global supply chain requiring neon gas, rare earth minerals, and highly specialized optics.

As economic historian Chris Miller argues in his book *Chip War*, semiconductors have effectively become the "new oil". Just as the 20th-century geopolitical landscape was defined by the Cold War and the Space Race, the 21st century is increasingly defined by a high-stakes technological standoff. This competition, however, is no longer limited to classical silicon. The "Chip War" has escalated into a race for Quantum Processing Units (QPUs). To illustrate this shift, we must look at the vast complexities of manufacturing quantum chips compared to classical architecture. While alternative architectures exist, the current industry standard relies on superconducting circuits. These utilize microscopic loops of superconducting metals that must be housed in cryogenic dilution refrigerators and cooled to approximately 15 millikelvin. We pointed to Google's release of the 105-qubit "Willow" chip as a prime indicator of U.S. leadership in this superconducting space. While possessing fewer active components than a traditional silicon processor, the Willow chip demonstrates drastically improved quantum coherence times, allowing it to execute specific calculations magnitudes faster than current supercomputers.

However, the hardware race is not a monolith. The landscape is becoming increasingly competitive as international players challenge the superconducting standard with entirely different physical modalities. A prominent example is the Canadian company Xanadu, which is pioneering room-temperature photonic circuits. By developing PennyLane - their open-source software library that integrates quantum computing with machine learning - Xanadu has positioned itself as a major contender in the global market. This implies a significant shift in the "war": companies are no longer just racing for more qubits, but are competing to discover which architecture can maintain a stable quantum state the longest without the massive infrastructure of cryogenic cooling. As these diverse

architectures emerge from different nations, it raises a critical question: as the industry pivots from silicon to a variety of competing quantum platforms, are we genuinely advancing global scientific capability, or are we simply resetting the board for a new era of monopolistic resource control?

Geopolitical Gatekeeping and Technological Sovereignty

We conceptualized the severe restrictions surrounding this hardware transition as "Geopolitical Gatekeeping." Our research outlined how controlling quantum hardware equates to controlling the future of advanced scientific discovery and artificial intelligence. At the foundational level, there is a material monopoly; China currently controls approximately 90% of the global gallium supply and 60% of germanium. In response, the U.S. and its allies have enforced severe manufacturing blockades, specifically banning the export of advanced extreme ultraviolet (EUV) lithography machines manufactured by ASML to rival nations.

We demonstrated that in the quantum sector, these blockades are even more acute. The cryogenic cooling systems required for superconducting QPUs depend on rare isotopes like Helium-3. Relying on M. Hawkins's 2024 analysis in Fortune Asia, we confirmed that the U.S. has enacted strict, targeted export controls specifically aimed at cryogenic systems, error-correction software, and these raw isotopes. To emphasize this point, fault-tolerant error-correction software is critical in quantum computing; because maintaining a delicate quantum state against environmental decoherence is immensely difficult, achieving reliable error correction has become a paramount objective for global competitors in the quantum race. Hawkins's reporting proves that the "Chip War" has officially expanded beyond silicon lithography. Consequently, we asserted that the development of quantum computing has transcended basic economics; it is now a legally enforced fight for technological sovereignty. This dynamic shatters the long-held tech industry illusion that infrastructure is built to make data "accessible to everyone." If the foundational hardware is legally cordoned off, we must critically ask: who actually gets to solve the world's most pressing problems? The ethics of Big Data can no longer solely focus on data privacy; they must fundamentally address the ethics of computational access.

The Astronomical Data Deluge and Classical Limits

Our presentation then bridged these geopolitical standoffs to their direct, severe implications for the field of astronomy, which is currently facing an unprecedented computational crisis. As we highlighted, modern sky surveys - such as the Rubin

Observatory's Legacy Survey of Space and Time (LSST) - will generate petabytes of data nightly, far outpacing the processing capabilities of standard classical hardware.

To establish the historical context of this crisis, we cited S. Kohler's 2018 publication in Eos, which outlines the immediate logistical nightmare observatories face as incoming high-resolution data outpaces standard analytical capabilities. We deepened this argument utilizing a 2010 decadal white paper by H.C. Ferguson et al. (submitted to the Astronomy and Astrophysics Decadal Survey). Ferguson's team accurately predicted that data reduction and analytical throughput would become the primary bottleneck of the next decade. Specifically, they noted that astronomical software systems were lagging behind hardware advancements because the community historically prioritized bespoke, instrument-specific programming. These siloed algorithms prevent the full utilization of available computational resources.

Due to these structural failures, classical models crash when attempting to process complex extraterrestrial environments. As noted by astrobiologist Caleb Scharf, while thousands of exoplanets are being discovered, classical supercomputers utilizing traditional distributed frameworks (like Hadoop and PySpark) lack the capability to efficiently scale the 3-D General Circulation Models required to accurately assess habitability. We argued that to overcome this "data deluge," the astronomical community must transition to Quantum Machine Learning (QML). However, the central paradox we identified is that deploying these advanced algorithms requires access to the exact QPUs currently restricted by export controls - presenting a frustrating irony for the scientific community. We are building massive, collaborative instruments like the LSST to democratize observational data, yet the computational power necessary to derive meaning from that data is held back by international restrictions. This forces us to question whether the future of astronomical discovery will be determined by the best scientific minds, or simply by the nations that monopolize the hardware.

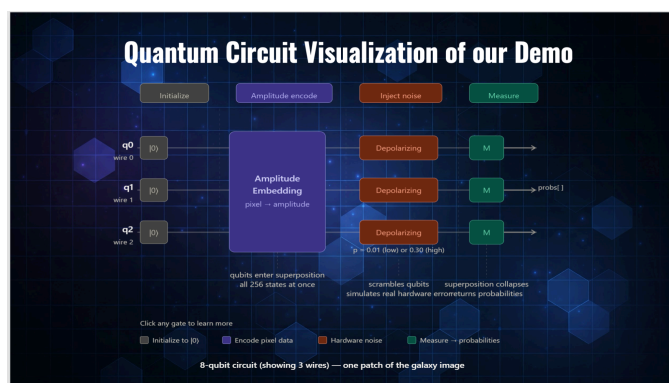


Figure 1: 8-Qubit Quantum Circuit Architecture for Image Reconstruction

This diagram outlines the technical layers of the QML algorithm utilized in our demonstration. It illustrates the workflow from the initial **State Preparation**(encoding 256 classical pixels into 8 qubits) through the **Amplitude Embedding** and **Quantum Variational Layers**, concluding with the **Measurement** phase that collapses the superposition to reconstruct high-fidelity astronomical imagery.

Quantum Mechanics and Circuit Architecture

To demystify how software interacts with this restricted hardware, we provided our audience with a technical breakdown of the quantum circuit architecture utilized for astronomical image processing. As discussed, classical computing relies on discrete bits existing in states of either 0 or 1, and quantum computing operates utilizing qubits. Qubits can exist in a state of 0, 1, or both simultaneously in a state of superposition. We used the analogy of a spinning coin to visualize this phenomenon: while a coin flat on a table exists in one of two states, a spinning coin exists as a blur of both, allowing quantum computers to process massive computational possibilities simultaneously.

To prove that these abstract physics concepts are actively being translated into programmable code, we referenced the 2024 IBM Qiskit documentation. The documentation acts as a software parallel to classical Big Data frameworks, demonstrating how quantum states can be manipulated programmatically to solve classification and regression problems. We explained that programming these quantum states requires applying quantum gates to manipulate, flip, and rotate the qubits. Furthermore, through entanglement, multiple qubits are linked to act as a single, synchronized system capable of holding massive amounts of data. We also detailed the vulnerabilities of these systems, specifically decoherence - how environmental interference causes qubits to lose their state mid-computation - before a final measurement forces the superposition to collapse back into discrete probability outputs. Understanding this architecture is crucial not just for programming, but for recognizing where the power lies: while the physical qubits are incredibly fragile and difficult to maintain, the mathematical frameworks we use to manipulate them are, paradoxically, highly robust and openly documented.

Technical Demonstration: PennyLane and QML Execution

We culminated our symposium with a live demonstration illustrating the practical application of QML on astronomical data, specifically reconstructing a high-resolution image of the Messier 51 (Whirlpool Galaxy). The empirical foundation for this demonstration was provided by Heredge et al.'s 2023 study in RAS Techniques and Instruments. Heredge's team successfully mapped an analytical workload of 40,000 galaxies onto a QPU for morphological classification using a quantum-enhanced support vector machine. Their research provides peer-reviewed proof that high-fidelity quantum hardware enables transformative efficiency for high-precision astrophysical tasks. This approach is further validated by Marchetti et al.'s 2025 review in EPJ Quantum Technology, which synthesizes recent literature to confirm that QML offers distinct computational advantages specifically for Earth observation and broad spatial data

processing. Building upon Heredge's methodology, our demonstration utilized PennyLane, a Python-based open-source library designed for quantum machine learning. We demonstrated how PennyLane acts as a crucial abstraction layer; its `@qml.qnode` decorator functions as a translator between traditional Python code and complex quantum operations. We walked the audience through our initial parameters, showing how the original galaxy image was segmented into 16x16 pixel patches. Because each patch contains 256 pixels, the system utilizes the mathematical rule of 2^n to establish the need for an 8-qubit circuit ($2^8 = 256$).

The core of the execution relies on PennyLane's Amplitude Embedding. Instead of manually calculating rotation angles, developers provide the numerical pixel data to the embedding function, which abstracts the complex mathematics required to shape the 8 qubits into a wave pattern whose probabilities perfectly encode the pixel values. To prove how the quality of the chip dictates scientific analysis, the demonstration utilizes PennyLane's default.mixed simulator. This allows the system to mimic a physical quantum chip operating on a classical CPU while artificially injecting a depolarizing channel to simulate environmental noise. The experiment acts as a comparative hardware reality check, stress-testing a highly restricted, high-noise QPU (30.0% decoherence rate) against an independent, high-precision QPU (1.0% decoherence rate). Finally, the `qml.probs()` function is called, collapsing the simulated quantum state and returning the outcome probabilities, which are mapped directly back to pixel brightness to reconstruct the galaxy image.

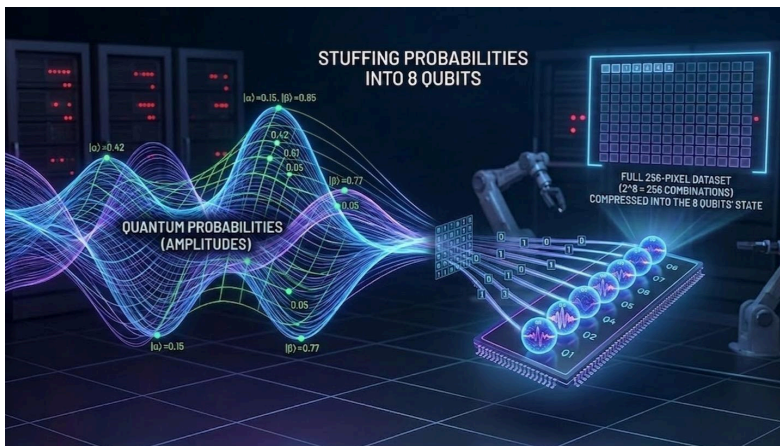


Figure 2: Amplitude Embedding and State Preparation

This visualization demonstrates the process of "stuffing" classical pixel probabilities into an 8-qubit quantum state. By utilizing superposition, the full 256-pixel dataset is encoded into the qubits' complex amplitudes, allowing the quantum circuit to process the entire image patch simultaneously through wave-pattern interference.

Ultimately, this live code walkthrough provided tangible evidence that hardware constraints fundamentally dictate the boundaries of scientific inquiry. If the global scientific community intends to utilize QML architectures to decode the universe, we must address the geopolitical infrastructure just as rigorously as the code itself. Moving forward, it is imperative to push for a future where access to revolutionary processing power is driven by scientific need rather than monopolized by sovereign borders.